FMO

Current Ratio-Case

PRUFROCK CORPORATION 2020 and 2021 Balance Sheets (in millions)	2020	2021	Change
Assets			
Current assets			
Cash	$ 84	$ 146	+$ 62
Accounts receivable	165	188	+ 23
Inventory	393	422	+ 29
Total	$ 642	$ 756	+$114
Fixed assets			
Net plant and equipment	$2,731	$2,880	+$149
Total assets	$3,373	$3,636	+$263
Liabilities and Owners' Equity			
Current liabilities			
Accounts payable	$ 312	$ 344	+$ 32
Notes payable	231	196	− 35
Total	$ 543	$ 540	−$ 3
Long-term debt	$ 531	$ 457	−$ 74
Owners' equity			
Common stock and paid-in surplus	$ 500	$ 550	+$ 50
Retained earnings	1,799	2,089	+ 290
Total	$2,299	$2,639	+$340
Total liabilities and owners' equity	$3,373	$3,636	+$263

PRUFROCK CORPORATION 2021 Income Statement (in millions)		
Sales		$2,311
Cost of goods sold		1,344
Depreciation		276
Earnings before interest and taxes		$ 691
Interest paid		141
Taxable income		$ 550
Taxes (21%)		116
Net income		$ 435
Dividends	$145	
Addition to retained earnings	290	

Current Ratio

2020 Current Ratio: $\frac{642}{543} = 1.18$

2021 Current Ratio: $\frac{756}{540} = 1.40$

Working Capital

	December 31, 2005
Current assets	$ 65,000
Current liabilities	(42,000)
Working capital	$ 23,000

Current Ratio

Current Ratio = Current Assets/ Current Liabilities

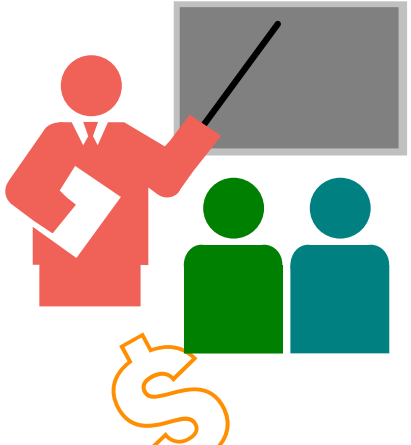
The current ratio measures a company's short-term debt paying ability.

A declining ratio may be a sign of deteriorating financial condition, or it might result from eliminating obsolete inventories.

Current Ratio

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

$$\text{Current Ratio} = \frac{\$65{,}000}{\$42{,}000} = 1.55$$



Quick or Acid Ratio

$$\frac{\textit{Cash} + \textit{Marketable Securities} + \textit{Accounts receivable}}{\textit{Current liabilities}}$$

Quick (or acid-test) ratio is computed just like the current ratio, except inventory is omitted since it is often the least liquid current asset.

- Relatively large inventories are often a sign of short-term trouble

Quick Ratio-Case

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Quick Ratio-Case

- 2020 Quick Ratio is 84+165/ 543= 0.46
- 2021 Quick Ratio 146+188/540=0.62

Quick Ratio-Case

- The quick ratio is a much more conservative measure than current ratio.
- This ratio measure the immediate solvency of the company.
- The ideal liquid ratio is 1:1. This is irrespective of nature of business.

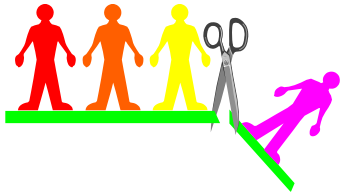
Acid-Test (Quick) Ratio

$$\text{Acid-Test Ratio} = \frac{\text{Quick Assets}}{\text{Current Liabilities}}$$

$$\text{Acid-Test Ratio} = \frac{\$50{,}000}{\$42{,}000} = 1.19$$

Quick assets include Cash, Marketable Securities, Accounts Receivable and current Notes Receivable.

This ratio measures a company's ability to meet obligations without having to liquidate inventory.



Cash Ratio-Case

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Cash Ratio

$$\frac{\textit{Cash + Marketable Securities}}{\textit{Current liabilities}}$$

Cash ratio may be of interest to a very short-term creditor.

- The cash ratio is a liquidity measure that shows a company's ability to cover its short-term obligations using only cash and cash equivalents.

Cash Ratio-Case

2020 Cash Ratio: $\frac{84}{543} = .15$

2021 Cash Ratio: $\frac{146}{540} = .27$

Liquidity Ratio

Liquidity ratios	Formula
Current ratio	(Current assets) / (Current liabilities)
Quick ratio	(Cash + Marketable securities + Receivables) / (Current liabilities)
Cash ratio	(Cash + Marketable securities) / (Current liabilities)

Ratio Analysis

Return Ratios

Return Ratio

The ratios that are of the most interest to stockholders include those ratios that focus on **net income, dividends, and stockholders' equities.**

NORTON CORPORATION 2023	
Number of common shares outstanding	
Beginning of year	17,000
End of year	27,400
Net income	$ 53,690
Stockholders' equity	
Beginning of year	1,80,000
End of year	2,34,390
Dividends per share	2
Dec. 31 market price per share	20
Interest expense	7,300
Total assets	
Beginning of year	3,00,000
End of year	3,46,390

Earnings Per Share

$$\text{Earnings per Share} = \frac{\text{Net Income} - \text{Preferred Dividends}}{\text{Average Number of Common Shares Outstanding}}$$

Whenever a ratio divides an income statement balance by a balance sheet balance, the average for the year is used in the denominator.

Earnings form the basis for dividend payments and future increases in the value of shares of stock.

Earnings Per Share

$$\text{Earnings per Share} = \frac{\text{Net Income} - \text{Preferred Dividends}}{\text{Average Number of Common Shares Outstanding}}$$

$$\text{Earnings per Share} = \frac{\$53{,}690 - \$0}{(\$17{,}000 + \$27{,}400)/2} = \$2.42$$

This measure indicates how much income was earned for each share of common stock outstanding.

Price-Earnings Ratio

$$\text{Price-Earnings Ratio} = \frac{\text{Market Price Per Share}}{\text{Earnings Per Share}}$$

$$\text{Price-Earnings Ratio} = \frac{\$20.00}{\$2.42} = 8.26 \text{ times}$$

A higher price-earnings ratio means that investors are willing to pay a premium for a company's stock because of optimistic future growth prospects.

Dividend Payout Ratio

$$\text{Dividend Payout Ratio} = \frac{\text{Dividends Per Share}}{\text{Earnings Per Share}}$$

$$\text{Dividend Payout Ratio} = \frac{\$2.00}{\$2.42} = 82.6\%$$

This ratio gauges the portion of current earnings being paid out in dividends. Investors seeking dividends (market price growth) would like this ratio to be large (small).

Return on Common Stockholders' Equity

$$\text{Return on Common Stockholders' Equity} = \frac{\text{Net Income} - \text{Preferred Dividends}}{\text{Average Stockholders' Equity}}$$

$$\text{Return on Common Stockholders' Equity} = \frac{\$53{,}690 - \$0}{(\$180{,}000 + \$234{,}390) \div 2} = 25.91\%$$

This measure indicates how well the company used the owners' investments to earn income.

Ratio Analysis

Capital Structure/Leverage Ratio

Leverage Ratio

These ratios indicate the mix of funds provided by owners and lenders.

Leverage ratios are of two types

a) Capital Structure ratios

b) Coverage ratios

Leverage Ratio

Capital Structure ratios:

These ratios provide an insight into the financing techniques used by a business and focus, as a consequence, on the long term solvency position.

Equity ratio:

This ratio indicates proportion of owners fund to total fund invested in the business.

Equity Ratio= Shareholder’s Equity/ Total Capital Employed

Leverage Ratio

Debt ratio:

Debt Ratio = Total Debt/ Capital Employed

Total debt includes short and long term borrowing from financial institution, debentures/ bonds deferred payment arrangements

Capital employed includes total debt and net worth.

This ratio is used to analyze long term solvency of the firm.

Leverage Ratio

Debt equity Ratio= Debt+ Preference Share Capital / Equity

This ratio indicates the proportion of debt fund in relation to equity. Lenders are very keen to know this ratio since it shows relative weight

- A high ratio here means less protection for creditors.
- A low ratio, on the other hand, indicates a wider safety cushion.

Coverage Ratio

Coverage ratios:

- The coverage ratios measure the firm’s ability to service the fixed liabilities.
- These ratios establish the relationship between fixed claims and what is normally available out of which these claims

Coverage Ratio

Debt Service Coverage ratio

$$\frac{\text{Earnings available for debt service}}{\text{Interest + Installments}}$$

Earnings for debt service= Net Profit + Non cash operating expenses like depreciation and other amortization

Lenders are interested in debt service coverage to judge the firms ability to pay off current interest and instalments.

Ratio Analysis

Activity Ratio

Activity Ratio

- Activity ratios are also called as **Turnover ratios or Performance** ratios.

- These ratios are used to evaluate efficiency with which the company manages and utilizes its assets.

- These ratios are usually calculated with reference to sale/cost of goods sold and are expressed in terms of rate or times. Some of the important activity ratios are as follows:

Activity Ratio

Capital Turnover Ratio=Sales/ Capital Employed

This ratio indicates the firm's ability of generating sales per rupee of long term investments.

Activity Ratio

Fixed Asset Turnover Ratio=Sales/Capital Asset

This ratio indicates the firm’s ability to efficient utilization of fixed asset in generating sales.

Activity Ratio

Working Capital Turnover Ratio= Sales/ Working Capital

Working Capital Turnover is further segregated into Inventory turnover, Debtors Turnover, Creditors turnover.

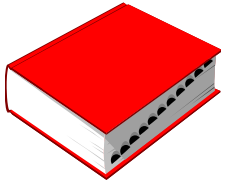
Ratio Analysis

Efficiency Ratio

Book Value Per Share

$$\text{Book Value per Share} = \frac{\text{Common Stockholders' Equity}}{\text{Number of Common Shares Outstanding}}$$

$$\text{Book Value per Share} = \frac{\$234{,}390}{27{,}400} = \$\ 8.55$$

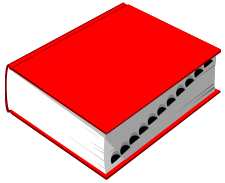


This ratio measures the amount that would be distributed to holders of each share of common stock if all assets were sold at their balance sheet carrying amounts and if all creditors were paid off.

Book Value Per Share

$$\text{Book Value per Share} = \frac{\text{Common Stockholders' Equity}}{\text{Number of Common Shares Outstanding}}$$

$$\text{Book Value per Share} = \frac{\$234{,}390}{27{,}400} = \$\ 8.55$$



Notice that the book value per share of $8.55 does not equal the market value per share of $20. This is because the market price reflects expectations about future earnings and dividends, whereas the book value per share is based on historical cost.

Ratio Analysis – The Short-Term Creditor

Short-term creditors, such as suppliers, want to be paid on time. Therefore, they focus on the company's cash flows and working capital.

NORTON CORPORATION 2005	
Cash	$ 30,000
Accounts receivable, net	
Beginning of year	17,000
End of year	20,000
Inventory	
Beginning of year	10,000
End of year	12,000
Total current assets	65,000
Total current liabilities	42,000
Sales on account	494,000
Cost of goods sold	140,000

Average Collection Period

$$\text{Average Collection Period} = \frac{365 \text{ Days}}{\text{Accounts Receivable Turnover}}$$

$$\text{Average Collection Period} = \frac{365 \text{ Days}}{26.7 \text{ Times}} = 13.67 \text{ days}$$

This ratio measures, on average, how many days it takes to collect an account receivable.

Ratio Analysis

Profitability Ratio

Profitability Ratio

Profitability Ratios:

Profitability ratios measure the profitability as a percentage of sales.

Net Profit Ratio =

Net Profit After Tax * 100/Sales

Profitability Ratio

Gross Profit Ratio = Gross Profit * 100/ Sales

Operating Profit Ratio = Operating Profit * 100/ Sales